Vision Document — eRegistrar

Course: CS425 — Software Engineering
Student: Ziad El Fatih — 618971
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1. Introduction

Course registration in the Computer Science department is still run on spreadsheets and email. The registrar builds each term's schedule by hand, faculty send their teaching preferences by email, and students register on paper forms that staff later key into the system.

This worked when the department was small. It now runs **four entries a year** with 100–130 students per entry, and offers 8–9 electives in a busy block. Faculty each have one or two specializations, a set of courses they can teach, and preferences about which blocks they are available in. Some 500-level courses have 400-level prerequisites, so the 400-level offerings must come in a student's earlier blocks. Holding all of this in a spreadsheet is slow, and prerequisite or capacity problems are usually discovered by students after registration has already opened.

eRegistrar is a web application that builds the term schedule with faculty assigned to each section, and lets students register online with the prerequisite and capacity rules enforced automatically.

2. Positioning

2.1 Problem Statement

The problem of	building the term schedule, assigning qualified faculty, and registering students for classes
Affects	the registrar, faculty, and students
The impact of which is	scheduling takes weeks of manual work, prerequisite and capacity violations are found late, and every schedule change forces a manual re-check
A successful solution would be	one system holding courses, faculty profiles and the schedule in a single database, applying the rules automatically, with web access for administrators, faculty and students

2.2 Product Position Statement

For	the registrar, faculty and students of the Computer Science department
Who	need a conflict-free term schedule and online registration against it
The eRegistrar	is a web-based course scheduling and registration system
That	generates the schedule from faculty qualifications and availability, and enforces prerequisites and capacity at the moment of registration
Unlike	spreadsheets plus email plus manual data entry
Our product	keeps one authoritative schedule, so a change made once is reflected everywhere and invalid registrations are rejected immediately

3. Stakeholders

Stakeholder	Role in the system
Registrar (Administrator)	Maintains courses and programs, generates and publishes the schedule, assigns faculty.
Faculty	Maintains their own profile — specializations, teachable courses, block availability — and views their assigned sections.
Student	Views the published schedule, registers for and drops courses.
Department Chair	Approves the published schedule; project sponsor.
IT Administrator	Deploys and operates the system, manages accounts.

User environment. Two to three registrar staff, about 25 faculty and up to 500 students on campus. All access is through a browser on desktop or phone; no software is installed on user machines. Registration load is concentrated in the first hour after the registration window opens.

4. Product Overview

4.1 Needs and Features — Problem / Need / Feature Table

#	Problem	Need	Priority	Feature
1	Faculty preferences arrive by email and are applied inconsistently	Faculty must own their qualifications and availability	High	**F1 — Faculty Profile Management**
2	Course and prerequisite data live in several spreadsheets that disagree	One authoritative catalog	High	**F2 — Course & Program Catalog**
3	Building a term schedule takes weeks of manual work	Generate the schedule automatically from the rules	High	**F3 — Schedule Generation**
4	Manual assignment double-books faculty or assigns courses they cannot teach	Detect assignment conflicts	High	**F4 — Faculty Assignment & Conflict Detection**
5	Students register on paper forms that staff re-key	Students register themselves with an immediate answer	High	**F5 — Online Student Registration**
6	Prerequisite and capacity violations are found after registration closes	Invalid registrations rejected as they are attempted	High	**F6 — Registration Rule Enforcement**
7	Nobody can see live enrolment	Role-specific views of the schedule and enrolment	Medium	**F7 — Schedule & Enrolment Views**
8	Anyone with the spreadsheet can change anything	Access must match the user's role	High	**F8 — Authentication & Role-Based Access**

4.2 Assumptions and Dependencies

1. The academic calendar of blocks and entries is fixed by the university.
2. Course and program definitions are maintained inside eRegistrar by the registrar.
3. Every user has a university account; authentication is delegated to the campus identity provider.
4. Deployment target is a Java/Spring Boot environment with a relational database.

4.3 Alternatives

- **Status quo (spreadsheets + email).** Free and familiar, but no rule enforcement, no concurrency control, and it does not scale to four entries a year.
- **Commercial SIS registration modules.** Comprehensive but priced and scoped for a whole institution, and they do not model this department's block/entry structure without heavy customization.
- **Build in-house — chosen.** The scheduling rules are specific to this department and are the hard part of the problem; a small focused application is cheaper than customizing a large product.

5. Other Product Requirements

Category	Requirement
Platform	Java 11+, Spring Boot, relational database, browser client
Performance	Pages respond within 2 seconds; 150 concurrent users during registration
Concurrency	Section capacity holds under simultaneous registration
Security	Role-based authorization on every operation; TLS; no passwords stored by the application
Usability	Student pages usable on a phone; staff productive after one walkthrough
Compliance	Student records handled in accordance with FERPA